AI Data Analytics Mini Project 1

Exploratory Data Analysis (EDA) Project Report

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Project Title

Video Game Sales (1980 - 2024) Analysis

Project Objectives

This report provides a comprehensive analysis of video game sales, both globally and regionally, from 1980 to 2024. By evaluating the impact of platform, genre and regions on sales performance, the study serves to provide game publishers and developers with actionable and data-driven recommendations for optimizing game development and marketing strategies.

Introduction to the dataset

The dataset was obtained from Kaggle. This study utilizes an extensive dataset consisting of over 64,000 game titles spanning more than 40 years of gaming history. By integrating global sales figures, regional market distributions (North America, Japan, Europe, Africa and other regions), and critical evaluations, the dataset bridges the gap between classic retro gaming and the modern era.

Dataset characteristics

The dataset comprises 14 features. The following table enumerates the variables and their variable types:

Variable Name	Description	Variable Type	Category	Data Type
img	URL for game cover	Qualitative	Nominal	Object
title	Game title	Qualitative	Nominal	Object
console	Platform that the game was released on	Qualitative	Nominal	Object
genre	Genre of the game	Qualitative	Nominal	Object
publisher	Publisher of the game	Qualitative	Nominal	Object
developer	Developer of the game	Qualitative	Nominal	Object
critic_score	Metacritic score (out of 10)	Quantitative	Continuous	Float64
total_sales	Global sales of copies in millions	Quantitative	Continuous	Float64
na_sales	North American sales of copies in millions	Quantitative	Continuous	Float64
jp_sales	Japanese sales of copies in millions	Quantitative	Continuous	Float64
pal_sales	European & African sales of copies in millions	Quantitative	Continuous	Float64
other_sales	Other regional sales of copies in millions	Quantitative	Continuous	Float64
release_date	Date that the game was released on	Temporal	N/A	Object
last_update	Date that the data was last updated	Temporal	N/A	Object

Data preprocessing and feature engineering

The initial dataset contains 14 columns and 64,016 rows:

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(64016, 14)
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The following feature engineering steps were performed:

(1) We deleted ‘img’ and ‘last_update’.
Rationale(s) behind: The ‘img’ is the URL for game cover on the website VGChartz, and the ‘last_update’ is the date where the data of that game was last updated. These two features were not required in the analysis so they were dropped.

(2) In 'developer', we found that there were some missing values, and we filled the missing values with 'Unknown'.
Rationale(s) behind: We did not remove the rows with missing values as valuable data might be lost. Instead, we filled the blank data with a new category called ‘Unknown’.

(3) In 'critic_score', we filled the missing values with the method of K-Nearest Neighbors (KNN) Imputation. The idea of KNN is to look at K similar rows to estimate what the missing value should be. Here we used the total sales and regional sales ('total_sales', 'na_sales', 'jp_sales', 'pal_sales' and 'other_sales') to help us approximate the missing scores, and we choose K=5.
Rationale(s) behind: Instead of just using the mean or median of a column to fill the missing values, we used KNN Imputation which was more accurate and reasonable. Using the data of sales number to estimate the missing data in game score implicitly implied that games of similar sales number had similar score.

(4) In 'total_sales', we deleted the rows with the value 0 and with missing values.
Rationale(s) behind: It is impossible for the total sales of a game to be zero. It was probably due to incorrect data entry. So we deleted the rows with the value 0. As for missing values, we found that rows with missing values in 'total sales' also had missing values in 'na_sales', 'jp_sales', 'pal_sales' and 'other_sales'. So we could not calculate the missing values in the total sales by summing up all the regional sales. For this reason, we removed the rows with missing values as well.

(5) In 'na_sales','jp_sales','pal_sales' and 'other_sales', we filled the missing values with 0.
Rationale(s) behind: Unlike total sales, the regional sales of a game can be zero, implying that the game was not released in that region, or the game was prohibited from being sold in that country due to the content rating system. So we replaced the missing values with 0 in these columns.

(6) In 'other_sales', there were some data in the format of date. We removed those rows and only retained the data which are integers or floating points.
Rationale(s) behind: It is impossible for a date to appear in the sales column, so we removed them.

(7) In 'release_date', we dropped those rows with missing values, then we extracted the year from the date.
Rationale(s) behind: For some old games, the exact release date might be unclear so the date was missing. As the release date of a game could not be estimated, we directly removed those rows with missing dates. And since the month and the day of a release date were not useful in our analysis, we only retained the year.

After data cleansing, the dataset contains 17,508 rows and 12 columns:

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(17508, 12)
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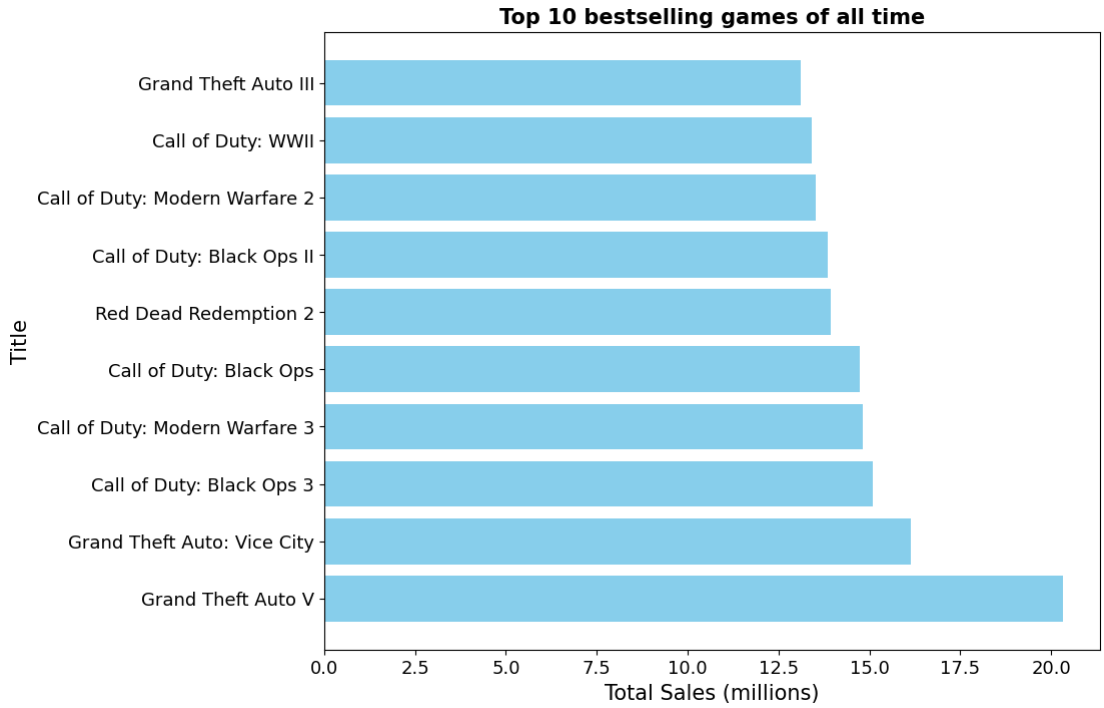
Introduction

Prior to the analysis, we first examine top ten highest-selling titles to understand historical market leaders:

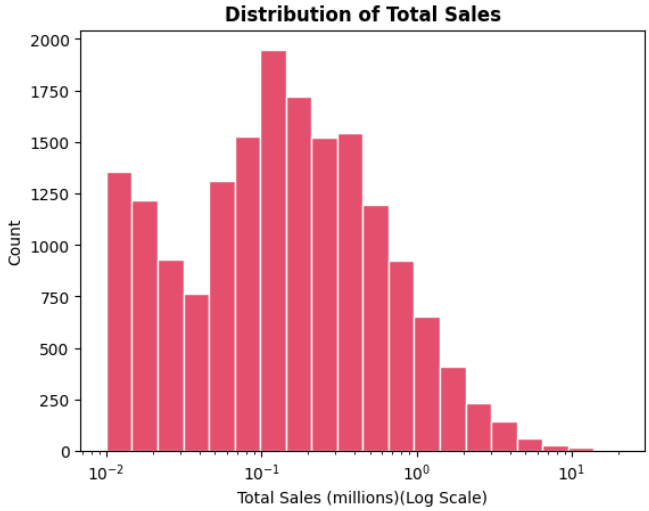


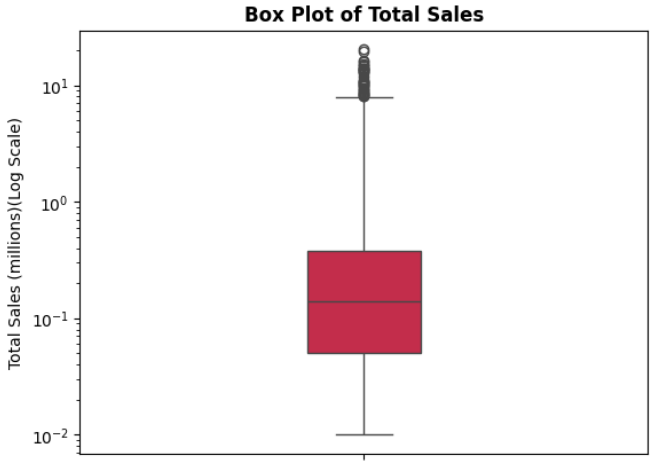
As can be seen from the above figure, the top 10 list is exclusively dominated by just three major intellectual properties (IPs): Grand Theft Auto, Call of Duty and Red Dead Redemption. We now apply various analytical techniques to extract actionable insights from the 40-year historical record.

Univariate Analysis

Total Sales

To assess the distribution of commercial success across the dataset, a histogram and a box plot were generated. A logarithmic scale was applied to the total sales to facilitate a clearer visualization of the market's density:





Key Observations:

(1) Histogram: The distribution exhibits a significant right-skewness under logarithmic transformation. This indicates a market structure where the vast majority of titles achieve modest sales, while the 'long tail' is composed of a small number of outliers

(2) Box Plot: The presence of numerous outliers confirms that total sales are not normally distributed. Instead, the industry is characterized by a "winner-takes-most" phenomenon, where peak performance is restricted to a small percentage of total releases.

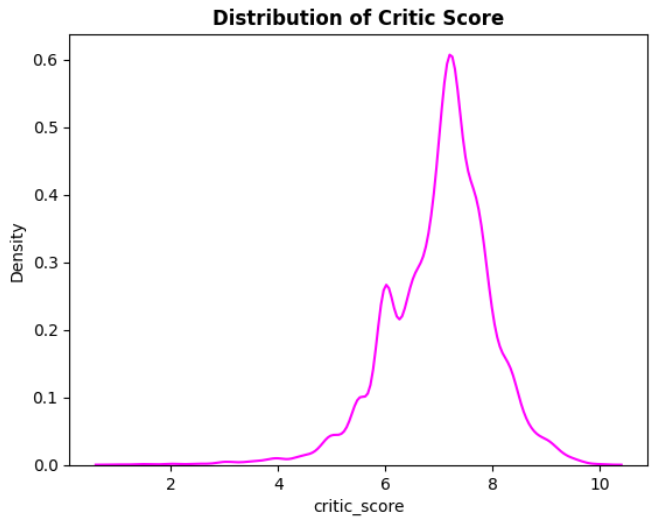
Strategic Insights:

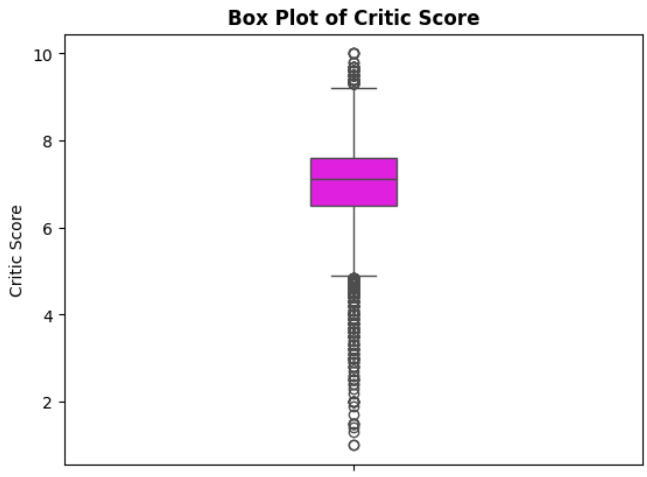
(1) Commercial success is highly consolidated within AAA Intellectual Properties (IPs), such as *Grand Theft Auto*, *Call of Duty*, and *FIFA*. These franchises function as market foundations, capturing a disproportionate share of global revenue.

(2) For major publishers—including Rockstar Games, Activision, and Electronic Arts—the data suggests that iterative development of current IPs remains the most reliable strategy for high returns. While these sequels may show variance in critical reception, the established brands provide a "revenue guarantee" that reduces the high financial risks associated with the development of new IPs.

Critic Score

To evaluate the distribution of critical score across the 40-year period, a Kernel Density Estimate (KDE) plot and a boxplot were utilized. These visualizations allow for assessment of how the industry's quality standards have shifted and concentrated over time:





Key Observations:

(1) KDE Plot: The distribution exhibits a slight left skewness, indicating that the majority of released titles receive favorable to moderate reviews. The data shows a median score of approximately 7.0 and a mode of 6.7, suggesting a strong concentration of scores within the mid-to-high range.

(2) Boxplot: The boxplot reveals significant dispersion in the form of bilateral outliers. While a substantial amount of titles fails to meet the lower limit (scoring below 5.0), a small portion of titles successfully penetrates the upper limit (scoring above 9.0).

Strategic Insights:

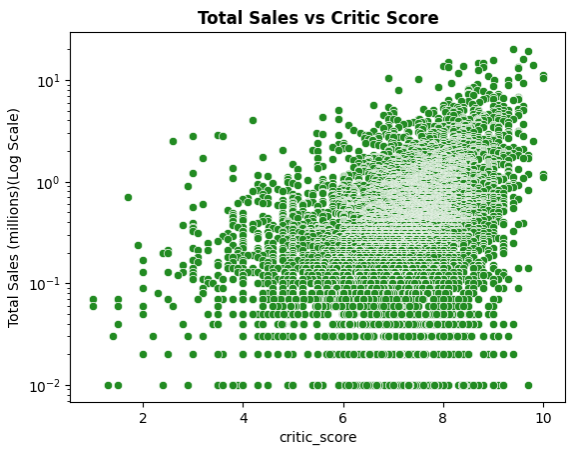
(1) A critic score of 7.0 represents the industry standard for the modern gaming industry. It suggests that a score of 7.0 is the minimum threshold for a title to be considered a standard commercial product.

(2) To minimize the risk of negative consumer perception, publishers and developers should adopt '7.0+' as a Key Performance Indicator (KPI) during the QA and development phases. Falling below this score significantly increases the probability of a title being classified as a 'lower-tier'.

Bivariate Analysis

Total sales vs critic score

To determine the extent to which public reviews drive commercial success, a scatter plot was generated to visualize the relationship between critic score and total sales:



A Pearson correlation coefficient (r) was also calculated to quantify the relationship between the two variables:

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Correlation =  0.33695185965566454
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Key Observations:

(1) The calculated correlation coefficient of r=0.337 indicates a positive but relatively weak relationship. While higher scores generally trend toward higher sales, the score alone is not an ultimate predictor of market performance.

(2) The scatter plot reveals significant variance in total sales, particularly within the 5.0 to 9.0 score range. In other words, games with identical critic scores exhibit vastly different commercial outcomes, suggesting that external factors—such as marketing spend, brand loyalty, and release timing—heavily influence the final sales figures.

(3) The data shows an absence of "breakout hits" below a critic score of 3.0. This suggests that while a high score does not guarantee success, a failing score acts as a barrier to high profits.

Strategic Insights:

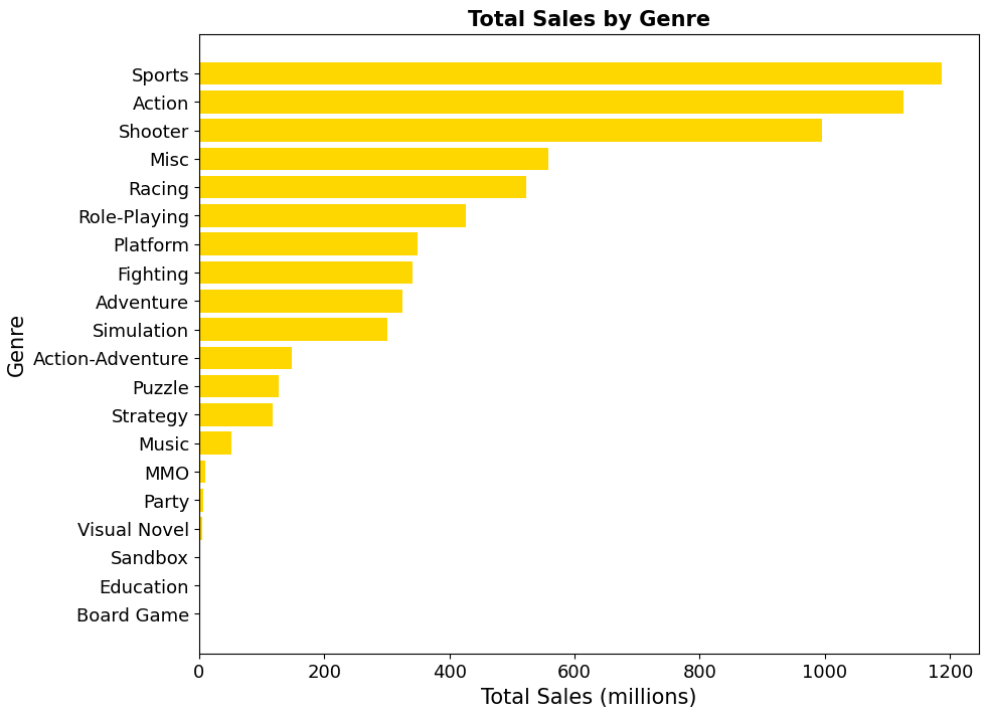
(1) A high critic score (>9.0) is not a must for high sales. Many prestige titles or indie games receive many positive reviews but fail to bring high revenue due to insufficient promotional budgets.

(2) The data confirms that becoming a million-seller almost requires a baseline of critical quality. Publishers should view high scores as a prerequisite for success rather than a guarantee of it.

(3) For developers, these findings highlight that a "perfect score" is less important for sales than the combination of decent quality (≥7.0) paired with aggressive marketing and platform visibility.

Gamers' preference by genre

The following section examines the distribution of total sales across various gaming genres over the last four decades. The horizontal bar chart illustrates the historical dominance of specific genres within the global market:



Key Observations:

(1) Historically, Sports titles represent the highest-selling genre, followed closely by Action and Shooter games. These three categories constitute the primary industry revenue.

(2) Genres typically associated with lower production budgets—such as Simulation, Puzzle, and Visual Novels—exhibit significantly lower cumulative sales compared to mainstream genres.

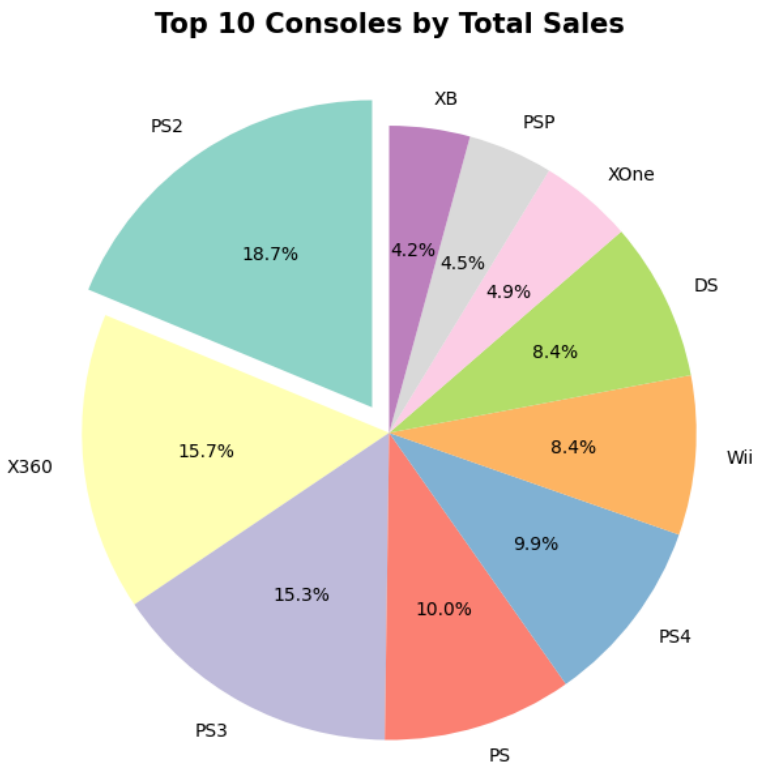
Strategic Insights:

(1) Based on historical market volume, stakeholders and developers should prioritize resources toward Sports, Action, and Shooter genres to maximize return on investment.

(2) Data suggests that genre is a more critical factor of commercial success than critical reception alone. A niche genre often has a lower "sales ceiling". For instance, an exceptionally high-quality Visual Novel is statistically unlikely to perform better than a mediocre Action title due to its limited market size.

Market share by consoles

The following analysis evaluates historical market distribution across consoles:



Key Observations:

(1) Sony constitutes the largest console market, with five consoles (PS2, PS3, PS, PS4, PSP) in the top ten. This indicates a "generational carry-over," where consumers consistently remain within the PlayStation ecosystem.

(2) The Xbox 360 (15.7%) and PS3 (15.3%) were nearly equal in market share. This era represents Microsoft’s peak competitiveness, driven largely by the success of the Xbox Live online service.

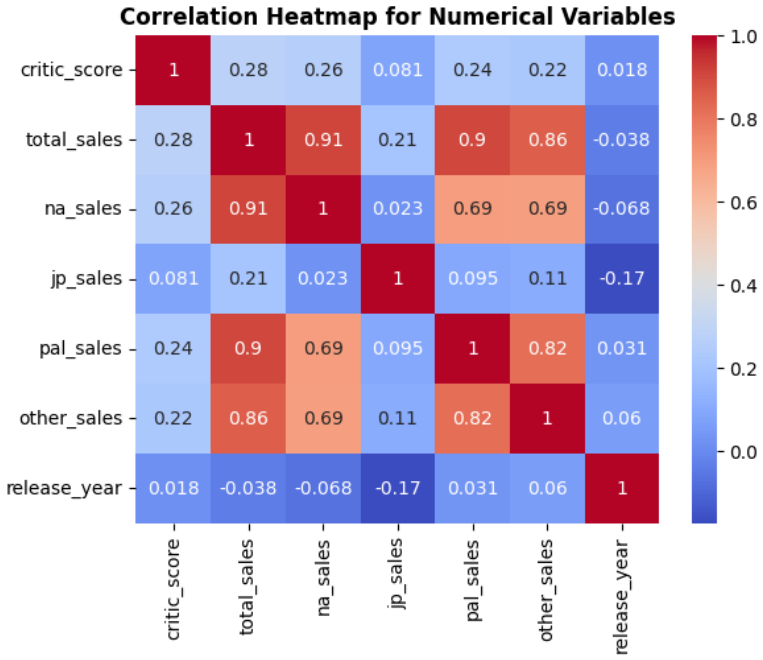
Strategic Insights:

(1) To mitigate risk, developers should avoid platform exclusivity. Targeting both PlayStation and Xbox grants access to over half the total market. This necessity drives the modern trend of porting former exclusive games to PC to maximise revenue.

(2) The historical success of the NDS and PSP highlights the importance of the handheld market. As casual players have migrated from portables to smartphones, developers must prioritize the mobile market to capture this significant audience.

Multivariate Analysis

The following heatmap analysis evaluates the relationships between regional sales, global sales, and critical reception to identify market trends:



Key Observations:

(1) Global sales show a near-perfect correlation with North American (0.91), European (0.90), and other regional markets (0.86).

(2) Japanese sales exhibit a significantly low correlation (0.21) with global trends, marking it a distinct and independent market.

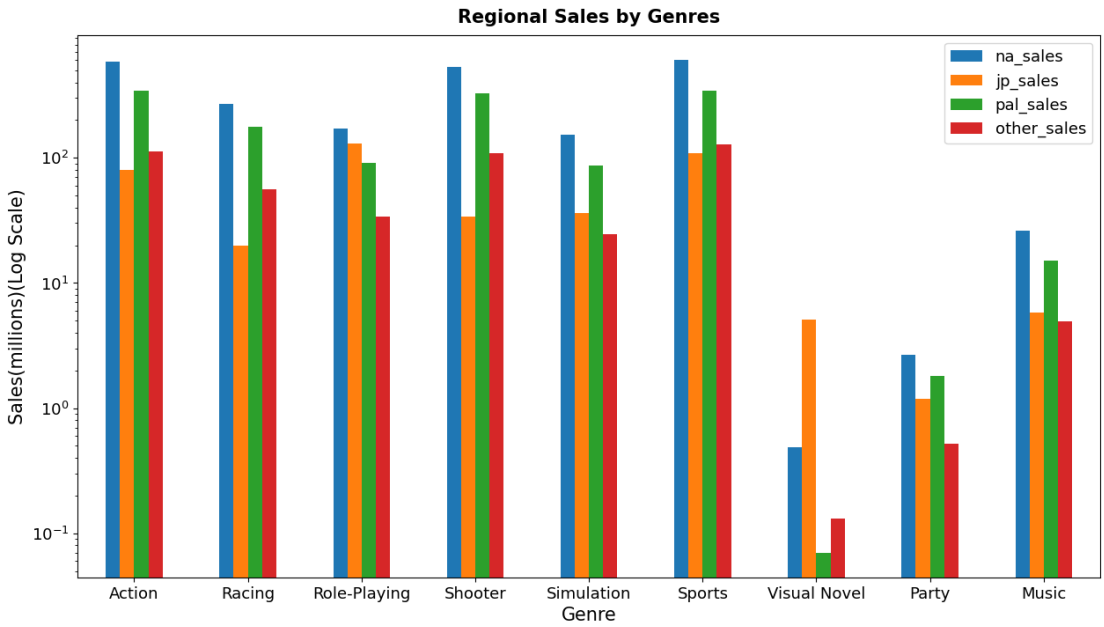
Strategic Insights:

(1) Due to its "unique" status, the Japanese market requires cultural adaptation such as character design and localised themes. Success in Western markets does not guarantee success in Japan, and vice versa.

(2) High correlation between Western regions suggests that a hit in North America is a reliable predictor for European success. Publishers should use North American pre-order data as a leading indicator to scale marketing and supply across other Western territories.

(3) To capture peak demand, AAA IPs should prioritize cross-platform releases and invest heavily on graphics and storytelling, as these elements drive success across most global regions.

The following bar chart further compares regional sales performance across popular global genres (Action, Shooter, Sports) and genres traditionally favored in the Japanese market (Role-Playing, Visual Novel):



Key Observations:

(1) High-performing global genres—specifically Action, Shooter, and Sports—exhibit their lowest relative sales volume within the Japanese market.

(2) Japan shows exceptionally high sales with Role-Playing Games (RPGs), with sales only slightly lower than North America. Notably, Japan is the primary market for Visual Novels, performing the best among other regions in this category.

Strategic Insights:

(1) Japan's preference for RPGs and Visual Novels over Western-style AAA games requires a distinct regional strategy.

(2) To penetrate the Japanese market efficiently, publishers should prioritize digital distribution to mitigate the cost associated with physical retail and logistics.

(3) The strong performance of Visual Novels and RPGs presents a high-potential entry point for indie developers. These genres allow for successful market capture with lower production budgets compared to AAA titles.

Challenges and Limitations

Challenges faced during the project

(1) "One Man Army": As I do not form group with other classmates, I need to do all the work myself, including searching for dataset, coding, performing data cleansing and analysis, writing report and giving presentation. I spend in average three hours per day on this project, which I feel exhausted and overwhelmed.

(2) Handling missing values and zeros in dataset: In the original dataset, there are many missing values and zeros. I tried to remove all rows which have missing values and zeros, but doing that would remain only about 1700 rows, which I think is not enough for data analysis, thus affecting the accuracy. In the end I decided to perform data cleansing column by column.

(3) Handling outliers: The 'total sales' column has a wide range of values, ranging from 0.01 to 20. When plotting the graphs involving the total sales, the great difference in values often results in difficult or unpleasant visualization. So I applied logarithmic scaling to these values, which enables easy and clear visualization.

Limitations of the approach or dataset

(1) Digital vs physical sales: It is unsure whether the dataset contains the digital sales data. In the modern era, digital downloads (Steam, Epic, PS Store) account for over 80% of sales. If the dataset lacks digital data, the total sales for newer consoles will be significantly underestimated.

(2) Survivor bias: The dataset only contains games that made it to market. It doesn't show the thousands of cancelled projects or games that sold zero copies, which limits my ability to analyze "failure factors".

Potential areas for future improvement

(1) Predictive modeling for hype vs reality: The current dataset looks at historical success. A powerful next step would be to correlate pre-release data with final sales. Pre-release data includes the number of likes from social media (Twitter, Instagram), YouTube trailer view counts, and Steam "Wishlist" numbers. The goal is to build a machine Learning model to see if "Social Sentiment" is a better predictor of success than "Critic Scores". This would help publishers decide where to allocate their marketing budget before a game launches.

(2) The impact of game subscription: With the rise of subscription service (Xbox Game Pass, PS Plus, EA Play), many users no longer buy individual games. Future analysis would be how being added to subscription service affects the total sales of a game. Would subscription lower the sales of a game because players can now access thousands of games through monthly fee? Can the total sales actually reflect the popularity of a game or we should count the number of downloads in subscription service instead? These are the questions to be investigated in future analysis.

Future Work

Ideas or approaches that were not implemented and the reasons

(1) Principal Component Analysis(PCA): PCA is a technique that can reduce the number of variables for easier data analysis. It involves finding the principal components that can represent variables of similar attributes. Since the implementation of PCA is quite complicated and time-consuming, I did not include it in my analysis.

(2) Random Forest: Random Forest is a method in machine learning that can predict or fill the missing values in a feature. In Random Forest model, many decision trees are built to form a 'forest'. Each decision tree contains criteria that are used to predict the result. The final result is then determined by majority voting (for categorical variables) or averaging(for numerical variables). In my analysis, I used KNN imputation to fill the missing values of the variable 'critic_score'. I only extracted the numerical features like total sales and sales in different regions to find the most similiar rows to approximate the missing scores. Random forest might be more accurate in approximating the missing scores since it can take categorical variables (console, genre, publisher etc) into account.

(3) Time Series Forcast: The dataset contains the variable 'release_year'. Time Series analysis could have been used to predict the future values of certain features, such as total sales and regional sales. But I did not include it in my analysis since it is not really accurate to predict future sales using the past data.

Conclusion

Summary of objectives and achievements

(1) This project aims to analyze the video game sales over 40 years of gaming history and provide recommendations and marketing strategies for game publishers and developers.

(2) In this analysis, we have looked into how the video game sales and the game scores distribute among all the games, the correlation between sales number and game score, the most popular genres over the years, the market share by different consoles, how the regional sales are related to the global sales and game preferences by different regions.

Key takeaways and insights

(1) Franchise Continuity: Top-tier IPs (e.g., *GTA, Call of Duty, FIFA*) account for over 90% of global sales. Maintaining these established franchises is a safe choice for publishers, as brand loyalty drives consistent sales even with few new elements.

(2) Quality Standard: While a high score does not guarantee a "hit", it acts as a commercial threshold. Acquiring million-plus sales typically requires a minimum review score of 7.0.

(3) Genre-First Development: Market volume is a primary success factor. Developers should target popular genres (Action, Shooter, Sports) to maximize returns, as niche genres often have low sales regardless of positive reviews.

(4) Cross-Platform Strategy: To capture over 50% of the console market, publishers must prioritize simultaneous multi-platform releases.

(5) Transition from Handheld to Mobile Market: The historical success of the NDS/PSP confirms a massive portable gaming audience. Developers should now focus on mobile platforms to capture the casual gamers market.

(6) Japanese Market Localization: To penetrate the Japanese market, Western developers must shift from standard AAA genres toward RPG and anime-inspired aesthetics to fit local cultural preferences.

(7) Enhancement of Indie Games Sales via Subscription: Small studios should join subscription services like Game Pass to mitigate financial risk, ensuring immediate revenue and broad exposure that compensates for lower traditional retail sales.

Reference

Divekar, B. (2024). Video Game Sales and Industry Data (1980-2024) [Data set]. Kaggle. https://www.kaggle.com/datasets/bhushandivekar/video-game-sales-and-industry-data-1980-2024